

11.3

$$2. f_x(2,1) \approx 3 \quad f_y(2,1) \approx 2$$

$$4. (a) f_x(-1,2) - \quad (b) f_y(-1,2) -$$

$$(c) f_{xx}(-1,2) + \quad (d) f_{yy}(-1,2) -$$

$$10. f_x = \frac{1}{2} x^{-\frac{1}{2}} \ln t \quad f_t = \frac{\sqrt{x}}{t}$$

$$13. \text{ for } x: \frac{ax+by}{cx+dy} = \frac{\frac{a}{c}(cx+dy) + (b - \frac{ad}{c})y}{cx+dy} \\ = \frac{a}{c} + \frac{(b - \frac{ad}{c})y}{cx+dy}$$

$$\frac{\partial f}{\partial x} = -1 (b - \frac{ad}{c})y (cx+dy)^{-2} c$$

$$= \frac{(ad-bc)y}{(cx+dy)^2}$$

$$\text{for } y: \frac{\partial f}{\partial x} = \frac{(bc-ad)x}{(cx+dy)^2}$$

$$17. R_p = \frac{1}{1+p^2 q^4} \quad q^2 = \frac{q^2}{1+p^2 q^4}$$

$$R_q = \frac{1}{1+p^2 q^4} \quad 2pq = \frac{2pq}{1+p^2 q^4}$$

$$20. \text{ assume } \int \sqrt{t^3+1} dt = \varphi(t) \quad \int_a^{\beta} \sqrt{t^3+1} dt = \varphi(\beta) - \varphi(a)$$

$$F_a = [\varphi(\beta) - \varphi(a)]' = -\varphi(a)' = -\sqrt{a^3+1}$$

$$F_\beta = [\varphi(\beta) - \varphi(a)]' = \varphi(\beta)' = \sqrt{\beta^3+1}$$

$$25. u = xy \sin^{-1}(yz)$$

$$\frac{\partial u}{\partial x} = y \sin^{-1}(yz)$$

$$\frac{\partial u}{\partial y} = x \sin^{-1}(yz) + xy \frac{z}{\sqrt{1-y^2 z^2}}$$

$$\frac{\partial u}{\partial z} = \frac{xy^2}{\sqrt{1-y^2 z^2}} \\ \frac{y}{z}$$

$$26. u = x^{\frac{y}{z}-1}$$

$$\frac{\partial u}{\partial x} = \frac{y}{z} x^{\frac{y}{z}-1}$$

$$\frac{\partial u}{\partial y} = (x^{\frac{1}{z}})^y = (x^{\frac{1}{z}})^y \ln x^{\frac{1}{z}} = \frac{1}{z} x^{\frac{y}{z}} \ln x$$

$$\frac{\partial u}{\partial z} = (x^y)^{\frac{1}{z}} = (x^y)^{\frac{1}{z}} \ln x^y \frac{-1}{z^2} = \frac{-y}{z^2} x^{\frac{y}{z}} \ln x$$

$$34. f_z = \frac{\sin z \cos z}{\sqrt{\sin^2 x + \sin^2 y + \sin^2 z}}$$

$$f_z(0,0,\frac{\pi}{4}) = \frac{\frac{\sqrt{2}}{2} \frac{\sqrt{2}}{2}}{\sqrt{0+0+\frac{1}{2}}} = \frac{\frac{1}{2}}{\frac{1}{\sqrt{2}}} = \frac{\sqrt{2}}{2}$$

$$40. F = x^2 - y^2 + z^2 - 2z - 4$$

$$\frac{\partial F}{\partial z} = 2z - 2 \quad \frac{\partial F}{\partial x} = 2x \quad \frac{\partial F}{\partial y} = -2y$$

$$\frac{\partial z}{\partial x} = \frac{\partial F}{\partial x} / \frac{\partial F}{\partial z} = \frac{2x}{2z-2} = \frac{x}{z-1}$$

$$\frac{\partial z}{\partial y} = \frac{\partial F}{\partial y} / \frac{\partial F}{\partial z} = \frac{-2y}{2z-2} = \frac{-y}{z-1}$$

$$44. (a) z = f(x)g(y)$$

$$\frac{\partial z}{\partial x} = g(y)f'(x) \quad \frac{\partial z}{\partial y} = f(x)g'(y)$$

$$(b) z = f(xy)$$

$$\frac{\partial z}{\partial x} = y f'(xy) \quad \frac{\partial z}{\partial y} = x f'(xy)$$

$$(c) z = f\left(\frac{x}{y}\right)$$

$$\frac{\partial z}{\partial x} = \frac{1}{y} f'\left(\frac{x}{y}\right) \quad \frac{\partial z}{\partial y} = \frac{-x}{y^2} f'\left(\frac{x}{y}\right)$$

$$47. w = \sqrt{u^2 + v^2}$$

$$\frac{\partial w}{\partial u} = \frac{u}{\sqrt{u^2 + v^2}} \quad \frac{\partial}{\partial u} \left( \frac{\partial w}{\partial u} \right) = \frac{\sqrt{u^2 + v^2} - u \frac{u}{\sqrt{u^2 + v^2}}}{u^2 + v^2}$$

$$= \frac{1}{\sqrt{u^2 + v^2}} - \frac{u^2}{(u^2 + v^2)^{\frac{3}{2}}}$$

$$\frac{\partial w}{\partial v} = \frac{1}{\sqrt{u^2 + v^2}} - \frac{v^2}{(u^2 + v^2)^{\frac{3}{2}}}$$

$$57. u = e^{r\theta} \sin \theta \quad \frac{\partial u}{\partial r} = \theta \sin \theta e^{r\theta}$$

$$\frac{\partial}{\partial r} \left( \frac{\partial u}{\partial r} \right) = \theta^2 e^{r\theta} \sin \theta$$

$$\frac{\partial}{\partial \theta} \left[ \frac{\partial}{\partial r} \left( \frac{\partial u}{\partial r} \right) \right] = (2\theta e^{r\theta} + \theta^2 r e^{r\theta}) \sin \theta + \theta^2 e^{r\theta} \cos \theta$$

$$60. g(x,y,z) = \sqrt{1+xz} - \sqrt{1-xy}$$

$$g_x = \frac{z}{2\sqrt{1+xz}} + \frac{y}{2\sqrt{1-xy}}$$

$$g_{xy} = \frac{2\sqrt{1-xy} + y \frac{y}{\sqrt{1-xy}}}{4(1-xy)}$$

$$g_{xyz} = 0$$

$$61. U_t = \sin kx (-a^2 k^2) e^{-a^2 k^2 t}$$

$$U_x = k e^{-a^2 k^2 t} \cos kx \quad U_{xx} = -k^2 e^{-a^2 k^2 t} \sin kx$$

$$a^2 U_{xx} = \sin kx (-a^2 k^2) e^{-a^2 k^2 t} = U_t$$

62.

(a)  $U_{xx} = 2$   $U_{yy} = 2$   $U_{xx} + U_{yy} \neq 0$  NO

(b)  $U_{xx} = 2$   $U_{yy} = -2$   $U_{xx} + U_{yy} = 0$  Yes

(c)  $U_{xx} = 6x$   $U_{yy} = 6x$   $U_{xx} + U_{yy} \neq 0$  NO

(d)  $U_{xx} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$   $U_{yy} = \frac{x^2 - y^2}{(x^2 + y^2)^2}$   $U_{xx} + U_{yy} = 0$  Yes

(e)  $U_{xx} = -\sin x \cosh y - \cos x \sinh y$

$U_{yy} = \sin x \cosh y + \cos x \sinh y$

$U_{xx} + U_{yy} = 0$  Yes

(f)  $U_x = -e^{-x} \cos y + e^{-y} \sin x$

$U_{xx} = e^{-x} \cos y + e^{-y} \cos x$

$U_{yy} = -e^{-x} \cos y - e^{-y} \cos x$

$U_{xx} + U_{yy} = 0$  Yes

64. (a)  $U_{tt} = -\sin(kx) a^2 k^2 \sin(akt)$

$U_{xx} = -\sin(akt) k^2 \sin(kx)$   $U_{tt} = a^2 U_{xx}$

(b)  $u = \frac{t}{a^2 t^2 - x^2}$   $U_x = \frac{2xt}{(a^2 t^2 - x^2)^2}$

$U_{xx} = \frac{2t(a^2 t^2 - x^2)^2 - 2xt(4x^3 - 4a^2 t^2 x)}{(a^2 t^2 - x^2)^4} = \frac{2a^4 t^5 - 12a^2 t^3 x^2 - 6tx^4}{(a^2 t^2 - x^2)^4}$

$U_t = \frac{-x^2 - a^2 t^2}{(a^2 t^2 - x^2)^3}$   $U_{tt} = \frac{-2a^2 t(a^2 t^2 - x^2)^2 - (-x^2 - a^2 t^2)(4x^3 - 4a^2 t^2 x)}{(a^2 t^2 - x^2)^4}$

$U_{tt} \neq a^2 U_{xx}$ , so isn't the solution

(c)  $U_{xx} = (x - at)^4 + (x + at)^4$

$U_{tt} = a^2(x - at)^4 + a^2(x + at)^4$   $U_{tt} = a^2 U_{xx}$  Yes

(d)  $U_{xx} = -\sin(x - at) + \frac{-1}{(x + at)^2}$

$U_{tt} = -a^2 \sin(x - at) + \frac{-a^2}{(x + at)^2}$

$U_{tt} = a^2 U_{xx}$ , so it is the solution

69.  $F = \frac{1}{R} - \frac{1}{R_1} - \frac{1}{R_2} - \frac{1}{R_3}$

$\frac{\partial F}{\partial R} = -\frac{1}{R^2}$   $\frac{\partial F}{\partial R_1} = \frac{1}{R_1^2}$   $\frac{\partial R}{\partial R_1} = \frac{\frac{1}{R_1^2}}{-\frac{1}{R^2}} = -\frac{R^2}{R_1^2}$

70. (a)  $P = \frac{mRT}{V}$   $\frac{\partial P}{\partial V} = -\frac{mRT}{V^2}$

$V = \frac{mRT}{P}$   $\frac{\partial V}{\partial T} = \frac{mR}{P}$

$T = \frac{PV}{mR}$   $\frac{\partial T}{\partial P} = \frac{V}{mR}$

$L = \frac{\partial P}{\partial V} \cdot \frac{\partial V}{\partial T} \cdot \frac{\partial T}{\partial P} = -\frac{mRT}{V^2} \cdot \frac{mR}{P} \cdot \frac{V}{mR}$

$L = -\frac{mRT}{VP}$

$\therefore PV = mRT \therefore L = -1$

(b)  $P = \frac{mRT}{V}$   $\frac{\partial P}{\partial T} = \frac{mR}{V}$

$\frac{\partial V}{\partial T} = \frac{mR}{P}$

$T \frac{\partial P}{\partial T} \cdot \frac{\partial V}{\partial T} = T \frac{mR}{V} \cdot \frac{mR}{P} = \frac{T m^2 R^2}{mRT} = mR$

75.  $f_{xy}(x, y) = 4$   $f_{yx} = 3$

$f_{xy} \neq f_{yx}$ , so can't believe

77. when  $y=2$

$4x^2 + 2y^2 + z^2 = 16 \Rightarrow 4x^2 + z^2 = 8$

$z = \sqrt{8 - 4x^2}$   $\frac{\partial z}{\partial x} = \frac{-8x}{2\sqrt{8 - 4x^2}}$  when  $x=1$

$\frac{\partial z}{\partial x} = -2 \therefore x=1-t$   $y=2$   $z=2+2t$

78. (a)  $\frac{\partial T}{\partial x} = -\lambda T_1 e^{-\lambda x} \sin(\omega t - \lambda x) + T_1 e^{-\lambda x} (-\lambda) \cos(\omega t - \lambda x)$

$= -\lambda T_1 e^{-\lambda x} [\sin(\omega t - \lambda x) + \cos(\omega t - \lambda x)]$

means the temperature attenuation rate

with the depth

(b)  $\frac{\partial T}{\partial t} = \omega T_1 e^{-\lambda x} \cos(\omega t - \lambda x)$

means the temperature attenuation rate

with the time

81. when  $y=0$   $f(x, y) = x^{-2}$

$f_x = -2x^{-3}$   $f_x(1, 0) = -2$